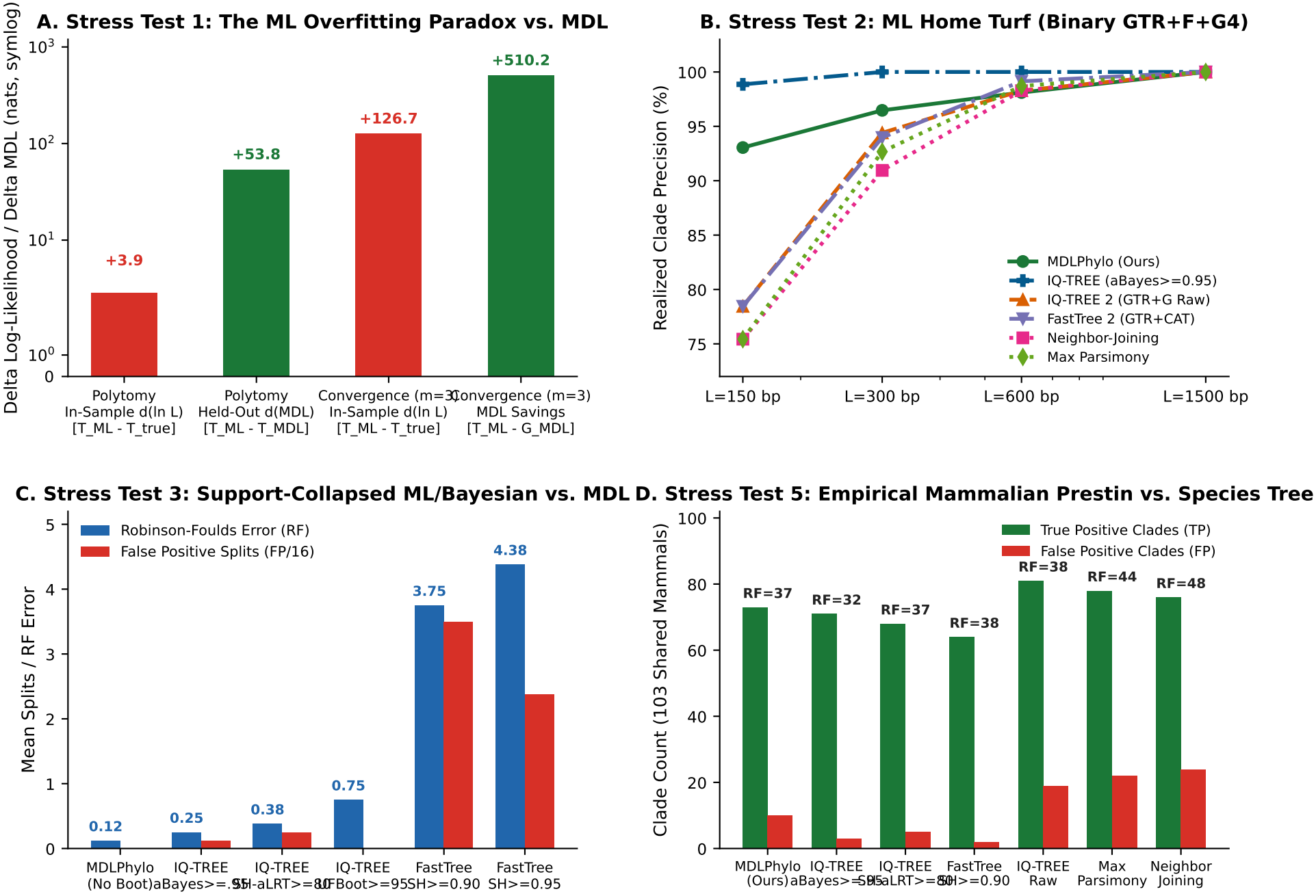
A. Stress Test 1: The ML Overfitting Paradox vs. MDL
Delta Log-Likelihood / Delta MDL (nats, symlog)
+3.9
+53.8
+126.7
+510.2
Polytomy In-Sample d(ln L) [T_ML - T_true]
Polytomy Held-Out d(MDL) [T_ML - T_MDL]
Convergence (m=3) In-Sample d(ln L) [T_ML - T_true]
Convergence (m=3) MDL Savings [T_ML - G_MDL]
B. Stress Test 2: ML Home Turf (Binary GTR+F+G4)
Realized Clade Precision (%)
L=150 bp
L=300 bp
L=600 bp
L=1500 bp
MDLPhylo (Ours)
IQ-TREE (aBayes>=0.95)
IQ-TREE 2 (GTR+G Raw)
FastTree 2 (GTR+CAT)
Neighbor-Joining
Max Parsimony
C. Stress Test 3: Support-Collapsed ML/Bayesian vs. MDL
Robinson-Foulds Error (RF)
False Positive Splits (FP/16)
Mean Splits / RF Error
0.12
0.25
0.38
0.75
3.75
4.38
MDLPhylo (No Boot)
IQ-TREE aBayes>=.95
IQ-TREE SH-aLRT>=80
IQ-TREE UFBoot>=95
FastTree SH>=0.90
FastTree SH>=0.95
D. Stress Test 5: Empirical Mammalian Prestin vs. Species Tree
True Positive Clades (TP)
False Positive Clades (FP)
Clade Count (103 Shared Mammals)
RF=37
RF=32
RF=37
RF=38
RF=38
RF=44
RF=48
MDLPhylo (Ours)
IQ-TREE aBayes>=.95
IQ-TREE SH-aLRT>=80
FastTree SH>=0.90
IQ-TREE Raw
Max Parsimony
Neighbor Joining